

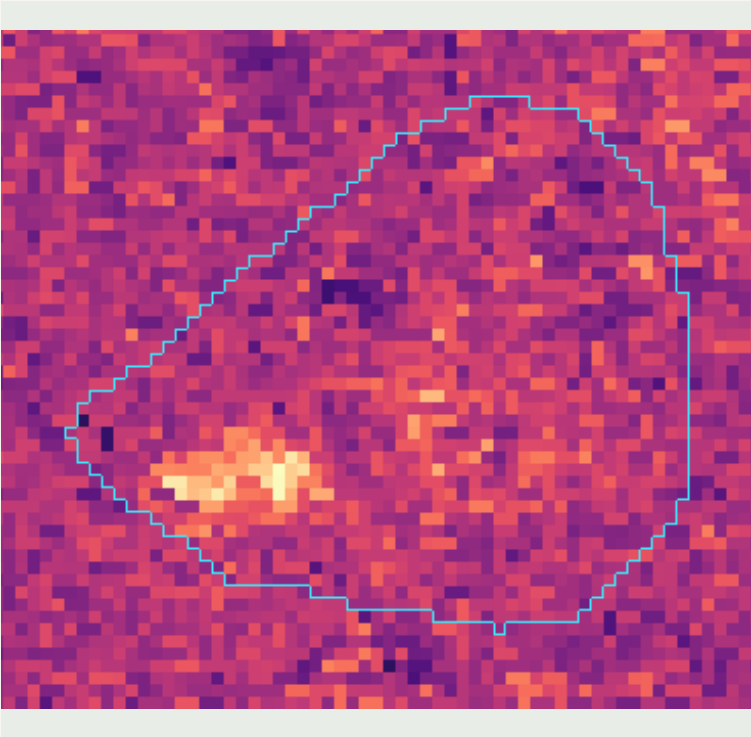
A new pair we can inspect

Volusia landfill, Florida | June 1, 2025 | Sentinel observed 12.7 minutes after EMIT

Sentinel-2 RGB



EMIT retrieval + provider mask



Model prediction

No prediction for this scene

Data checks must pass before a compatible model is trained or evaluated here.

What I have now

I can place the provider's EMIT plume annotation over actual Sentinel imagery and inspect the retrieval. The full annotation fits inside the crop. The five target bands are downloaded; the two missing SWIR bands were completed in a bounded 2.75 MiB pass.

This is a data example, not a model result. The reference is natively 60 m. A finer display does not create finer measured truth. Label-zero semantics, retrieval units and radiometric conversion still need verification; a temporal reference and paired negative are also missing.

Source: UNEP-IMEO / MARS-Hyperspectral-EMIT-v2025 (CC-BY-NC-SA-4.0); Sentinel-2 via Earth Search. Source links and receipts accompany the offline review.

A localized result does not establish a working detector

Saved R5 mean prediction | November 1, 2023 | Fixed threshold 0.50 | Reused development data

Sentinel-2 RGB



Reviewed reference



Saved model output



This example overlaps part of the reviewed plume

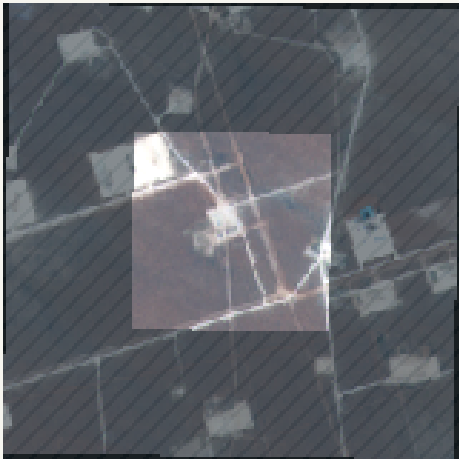
On the supported central area, this scene has 38.8% intersection-over-union (IoU). But the full 65-scene development run has only 2.15% pooled IoU and 2.25% precision. The run failed its quality gate. These images are diagnostic examples, not an estimate of general performance.

Cyan: reviewed reference boundary. Orange/red: saved positive prediction. Dimmed area: no model output or no valid comparison. Only the central 88 x 88 pixels of the 200 x 200 source are supported. The reference here is a MARS reviewed mask, not quantitative EMIT enhancement.

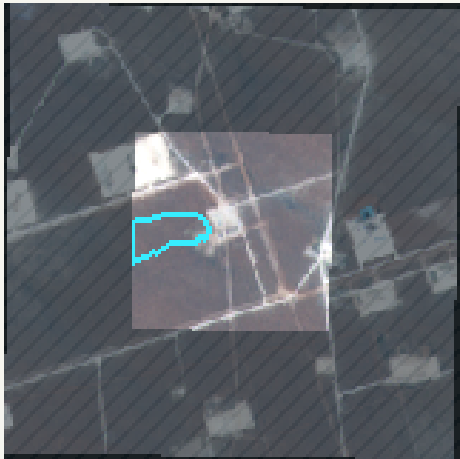
The misses and false positives matter

Both examples use the same saved model and cutoff. The interactive demo includes all three positives and two largest false-positive controls.

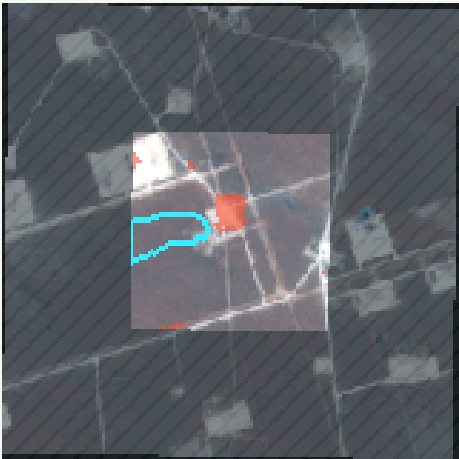
Missed plume



Sentinel-2 RGB



Reviewed reference



Saved model output

Largest false-positive control



Sentinel-2 RGB



Reviewed reference



Saved model output

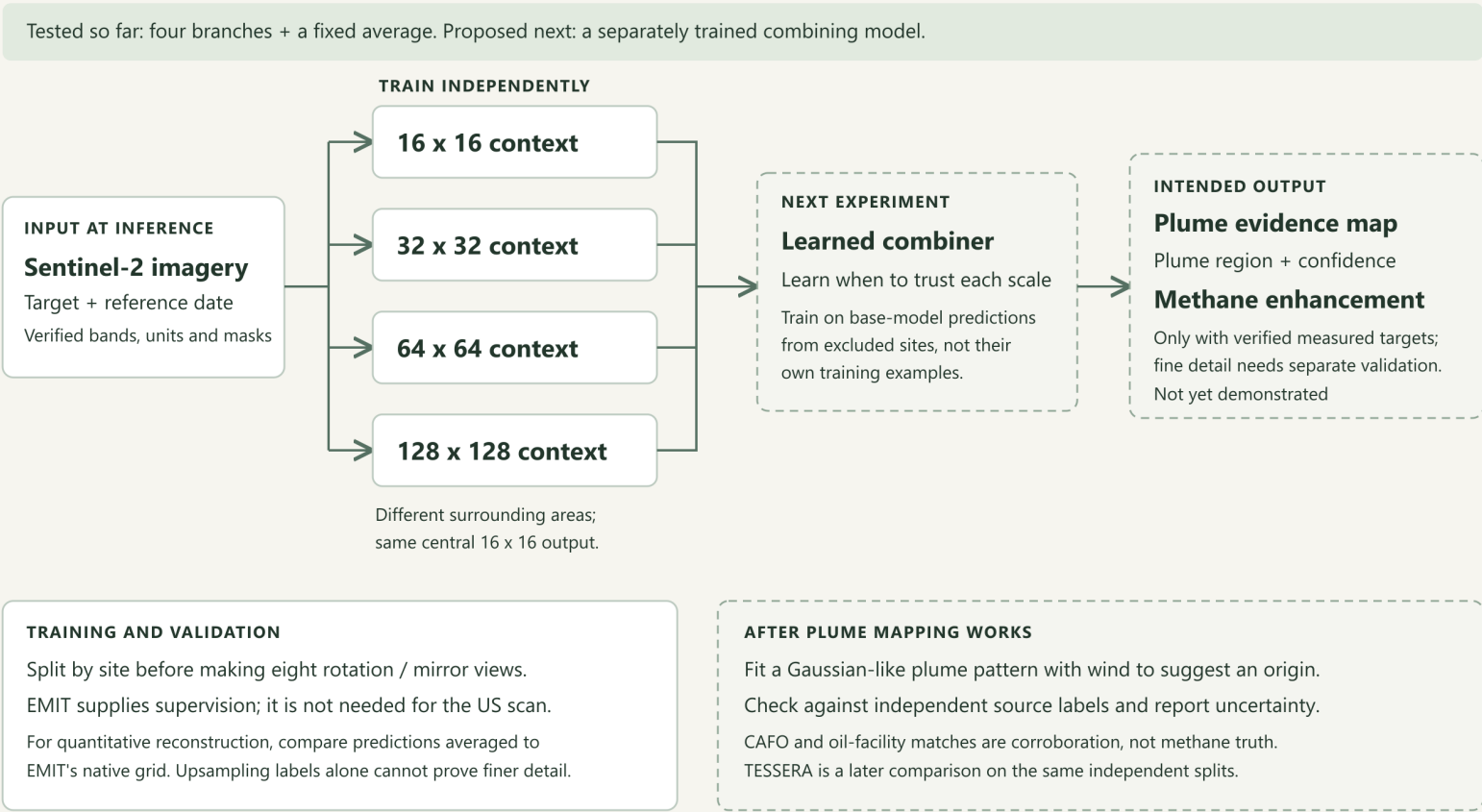
The missed scene recovers 0 of 461 labeled plume pixels. The no-plume control activates 869 of 7,744 evaluated pixels. These failures are why visual checks and reviewed controls are required before another training run or a US-wide scan.

Martin's architecture and the decisions we need

The US screening goal stays the same. The next milestone needs trustworthy supervision and a clear acceptance test.

Martin's multi-scale model

Nationwide goal: use Sentinel imagery to flag possible methane plumes for review.



Solid = available components or evaluation rules. Dashed = proposed capability. Existing R5 results failed the quality gate.

Agree on the first success criterion

Start with reliable plume regions and controlled false alarms; retain finer methane reconstruction as a separately validated goal.

Obtain the label definitions

Can Martin connect us to the provider or exact processing documentation for mask zero, retrieval units, quality flags and reviewed negatives?

Validate source locations later

Request independent source and wind information for the Gaussian-like origin fit. A landfill or CAFO coordinate alone is not source truth.

My next experiment

After data checks pass: split by site, create eight rotation/mirror training views, train four scales independently, then compare a fixed average with a combiner trained on predictions from excluded sites. Review every positive, every miss, and the largest false positives before expanding.